



Tiny robots discover learning process from ants

Ants are found to guide other ants to the new location of the nest. One ant learns the route and initiates the run. It teaches another ant, which teaches another, and the process goes on. Researchers at Bristol's School of Biological Sciences designed a tiny robot with similar scent glands for the ants and placed it in an arena with worker ants consisting of an ant nest and a new location to relocate. The robot initiated the run and provided a positive outcome as it successfully guided other ants to new locations and even back. This method of learning and teaching algorithms can improve the efficiency of an autonomous system.

Plastics for use in space being researched

Plastics have amazing properties that can be useful even in space technology. The only limitation is that it cannot sustain extreme radiation, which is the only reason that it may not be used in space. A PhD student from Florida is doing research to develop strong plastics with higher thermal stability that can withstand high temperatures and radiation in space. She is working on a method to combine nanomaterials and plastic in a controlled, consistent way.



Ortikasa performs a thermal conductivity measurement of a composite (Credit: <https://news.fiu.edu>)

In the traditional method, the mixing of nanomaterial with plastic is a big task that creates clumps. If this research is successful, it may revolutionise space technology.

Spider silk helps study biological solutions

Studies at Taiwan's National Yang Ming Chiao Tung University have looked at how the refractive index of biological solutions changes using spider silk. They made a sensor using harnessed spider silk from *Nephila pilipes*, which is 10 microns in diameter. The silk obtained is spun with 100-micron optic fibre that consists of biocompatible resin. A thin layer of biocompatible nano gold is added to

improve its efficiency. Lastly, the researchers immersed one end in the liquid sample and the other end in the light source and spectrometer. It enabled the researchers to detect the refractive index and to determine the sugar type and the concentration. The team is further developing software to enable the sensor to be used in mobile devices for card readings.

The 'space plate' can further miniaturise cameras, telescopes, etc

Scientists at Cornell led by doctoral student Kunal Shastri and assistant professor Francesco Monticone have defined the ultimate practical and fundamental limits to space compression via the use of a 'space plate' to reduce and remove the amount of free space, which usually means the space between the sensor and lens needed for light to better focus. The space plate imitates the response of the free space at a much lesser length. It has a wide range of applications, including cameras. It has the potential to miniaturise telescopes, projectors, and even the electromagnetic spectrum using antennae.

Connecting thousands of IoT devices just got easier

The creation of an mmWave Backscatter System by a team from KAIST School of Electrical Engineering is great news for the IoT market since it will be able to offer quick and dependable connectivity even for a large network, which



The research was nominated Best Paper Award at ACM MobiSys 2022 (Credit: KAIST)

may allow IoT devices to finally realise their full potential. This is a system that enables simultaneous signal demodulation in a complicated communication environment where tens of thousands of IoT devices are organised indoors using their mmWave backscatter approach. mmWave has a large frequency range that exceeds 10GHz, which offers excellent scalability. Also, the designed system is easily adaptable since it can use the radar of already-available commercial goods as a gateway.